Network Virtualization Technologies – Summary

1. VLAN (Virtual LAN)

- Splits one physical network into multiple logical networks.

- Devices communicate as if on same LAN.

- Uses 802.1Q VLAN tagging.

Basic Config (Cisco):

- Create VLAN: vlan 10 → name Students

- Assign port: interface fa0/1 → switchport mode access → switchport access vlan 10

- Trunk port: interface fa0/24 → switchport mode trunk

2. VPN (Virtual Private Network)

- Encrypted tunnels over internet.

- Uses IPsec / SSL / GRE.

- Supports remote access and site-to-site.

Basic IPsec Steps:

1. ISAKMP policy

2. Transform-set

3. Crypto map

4. Apply to interface

3. Virtual Switch (vSwitch)

- Software switch in hypervisors (VMware/Hyper-V/KVM).

- Connects VMs internally and to external network.

4. SDN (Software-Defined Networking)

- Control plane centralized in SDN controller.

- Switches follow OpenFlow rules.

- Enables programmable, automated networks.

5. NFV (Network Function Virtualization)

- Virtual appliances replace hardware firewalls/routers.

- Runs as VM or container.

6. VXLAN

- Encapsulates L2 in UDP.

- Supports multi-site overlays.

- 16 million VNIs.

7. Container Networking

- Uses veth pairs, Linux bridges, overlays.

- Kubernetes uses CNI plugins (Calico/Flannel).